Evaluating Feature Attribution Stability Using SHAP and LIME Across Multiple Machine Learning Models

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Abstract

We train three tree-based models (Random Forest, Gradient Boosting, and XGBoost) on the same tabular datasets and compare their SHAP-based and LIME-based global feature importance and *instance-level* (local) stability. Using three public datasets—California Housing, Adult income, and Titanic—we compute pairwise Spearman rank correlation of feature importance (SHAP and LIME), top-k overlap, per-instance stability, and *SHAP–LIME* rank correlation per model. We find that SHAP-derived rankings are often highly correlated across models but local attributions are less stable; LIME rankings can be stable across models yet disagree with SHAP. We report test-set performance and recommend checking consistency across models and across explanation methods (SHAP vs. LIME).

Keywords: SHAP; LIME; feature attribution; explainable AI; model comparison; stability; tree ensembles.

1 Introduction

SHAP (SHapley Additive exPlanations) is widely used to explain predictions of machine learning models by attributing importance to each input feature. When multiple models achieve similar predictive performance on the same task, a natural question is whether their explanations agree: do they identify the same features as important, and in the same order?

Prior work has shown that feature attributions can vary with model choice and that different models with comparable accuracy may yield different local explanations. We quantify this phenomenon for *global* SHAP-based feature importance across three popular tree-based models—Random Forest, Gradient Boosting, and XGBoost—on three public datasets. We also address the gap of *instance-level* stability: whether local attributions are stable across models for the same instance, and how to summarize that (e.g., per-instance rank correlation, proportion of "stable" instances). We measure agreement via pairwise Spearman rank correlation and top-k overlap. We also compare SHAP with LIME (Local Interpretable Model-agnostic Explanations): both are post-hoc attribution methods, but LIME fits a local linear surrogate while SHAP uses Shapley values; we report whether their global importance rankings agree per model.

2 Related Work

Consistency of SHAP within a single model (e.g., tree ensembles) is well studied. The Rashomon set framework addresses explanation variability across models that perform equally well: different models in the set can yield different attributions for the same prediction. SHAP rankings have also been shown to be sensitive to architecture and random initialization in deep learning. We focus on tabular data and tree explainers, and provide a systematic empirical comparison with explicit metrics (Spearman, top-k overlap) that can be replicated.

3 Methods

3.1 Datasets and Labels

We use three public datasets:

- **California Housing**: regression; target is median house value (MedHouseVal). Source: sklearn `fetch_california_housing`.
- **Adult**: classification; target is income > 50K (binary). Source: OpenML adult, version 2. Numeric features only.
- **Titanic**: classification; target is survival (binary). Source: OpenML titanic, version 1.

Data are split 80%/20% train/test with a fixed random seed (42) for reproducibility.

3.2 Models

For each dataset we train three models with default or standard settings ($n_estimators = 100$, fixed `random_state`=42):

- Random Forest (scikit-learn)
- Gradient Boosting (scikit-learn)
- XGBoost

Regression models are used for California Housing; classifiers for Adult and Titanic.

3.3 SHAP and Global Importance

We use TreeExplainer (SHAP) with a small background set (100 samples from the training set) and compute SHAP values for a subsample of the test set (200 samples by default). For each model we define *global* feature importance as the mean absolute SHAP value per feature over this subsample. Features are then ranked by this importance (rank 1 = most important).

3.4 LIME

We use LIME (LimeTabularExplainer) on the same subsample of instances. For each instance we fit a local linear surrogate (default 1000 samples) and obtain feature weights; we aggregate to *global* LIME importance as the mean absolute weight per feature and rank features accordingly. LIME is model-agnostic and can yield different rankings than SHAP. We compute LIME-vs-LIME stability (pairwise Spearman and top-k overlap across the three models) and SHAP–LIME rank correlation per model. LIME can be disabled with `-no-lime` for faster runs.

3.5 Stability Metrics

- **Spearman rank correlation**: pairwise, between the feature rankings of each pair of models. High r indicates similar ordering of importance.
- **Top-k overlap (Jaccard)**: for each pair of models, the Jaccard index of the sets of top-k feature indices ($k = 5$). Equal to 1 if the same k features are selected (order ignored).

3.6 Instance-level (local) stability

Global metrics (mean |`SHAP`|) do not capture whether *per-instance* attributions agree across models. For each instance in the SHAP subsample we compute: (i) pairwise Spearman correlation of the three models' SHAP vectors (across features), then the mean pairwise ρ over the three pairs; (ii) for each pair, the top-k feature set from that instance's SHAP values, and the Jaccard index between those sets—we take the minimum Jaccard over the three pairs (worst agreement). We summarize across instances by the mean and standard deviation of (i), the *proportion of instances* with mean pairwise $\rho \geq 0.9$ ("Spearman-stable"), the mean of (ii), and the proportion of instances for which all three models have identical top-k sets ("top-k-stable"). This addresses the gap of whether local attributions are stable across models for the same instance.

3.7 Model Performance

We report test-set metrics to contextualize explanation comparison: RMSE and R^2 for regression; accuracy and AUC for classification.

4 Results

All numbers below are from a single run with `random_state=42` and `-shap-sample 200`. Reproduce with: `python shap_stability.py -dataset california` (and `adult`, `titanic`).

Table **??** summarizes model performance. Table **??** gives pairwise Spearman correlation of SHAP feature rankings; Table **??** gives top-5 Jaccard overlap. Table **??** summarizes instance-level stability (mean pairwise Spearman per instance, proportion Spearman-stable, mean min top-5 Jaccard, proportion top-5 identical). Table **??** shows the top-5 features by mean |`SHAP`| per model for California Housing. Figure **??** illustrates global feature importance for one dataset. Rankings are often highly correlated but the top-5 set or order can differ across models (e.g., XGBoost vs. others on California); local stability is generally lower than global. When LIME is run on the same instances, LIME-vs-LIME Spearman is often high (models agree under LIME), but SHAP–LIME Spearman per model can be low or negative, indicating that the two methods do not always identify the same features as important.

Table 1: Test-set performance (single run, seed 42).

Dataset	Model	Metric 1	Metric 2
California	RF	RMSE 0.505	R^2 0.805
	GBM	RMSE 0.542	R^2 0.776
	XGB	RMSE 0.472	R^2 0.830
Adult	RF	Acc. 0.809	AUC 0.827
	GBM	Acc. 0.844	AUC 0.868
	XGB	Acc. 0.850	AUC 0.873
Titanic	RF	Acc. 0.798	AUC 0.860
	GBM	Acc. 0.782	AUC 0.876
	XGB	Acc. 0.809	AUC 0.862

5 Discussion

Even when model performance is similar (e.g., all three classifiers on Titanic around 78–80% accuracy), the ordering or composition of the top SHAP features can differ. SHAP and LIME can also disagree on which features matter most, so conclusions should not rely on a single explanation

Table 2: Pairwise Spearman correlation of SHAP feature rankings (single run).

Dataset	Model pair	Spearman r
California	RF–GBM	0.929
	RF–XGB	0.881
	GBM–XGB	0.952
Adult	RF–GBM	0.886
	RF–XGB	0.771
	GBM–XGB	0.943
Titanic	RF–GBM	0.964
	RF–XGB	0.929
	GBM–XGB	0.964

Table 3: Top-5 feature overlap (Jaccard index) between model pairs.

Dataset	Model pair	Jaccard (top-5)
California	RF–GBM	1.000
	RF–XGB	0.667
	GBM–XGB	0.667
Adult	RF–GBM	0.667
	RF–XGB	0.667
	GBM–XGB	1.000
Titanic	RF–GBM	1.000
	RF–XGB	1.000
	GBM–XGB	1.000

Table 4: Top-5 features by mean |`SHAP`| per model (California Housing).

Model	Top-5 features (most to least important)
RF	MedInc, Latitude, Longitude, AveOccup, HouseAge
GBM	Latitude, Longitude, MedInc, AveOccup, HouseAge
XGB	Latitude, Longitude, MedInc, AveOccup, AveRooms

Table 5: Instance-level (local) stability: per-instance agreement across the three models (single run, 100 instances). Mean ρ = mean over instances of mean pairwise Spearman; Prop. stable = fraction of instances with mean pairwise $\rho \geq 0.9$; Mean min Jaccard = mean over instances of minimum pairwise top-5 Jaccard; Prop. top-5 identical = fraction of instances where all three models have the same top-5 set.

Dataset	Mean ρ (std)	Prop. stable	Mean min Jaccard	Prop. top-5 identical
California	0.79 (0.11)	15%	0.66	17%
Adult	0.67 (0.23)	10%	0.71	14%
Titanic	0.73 (0.15)	7%	0.66	19%

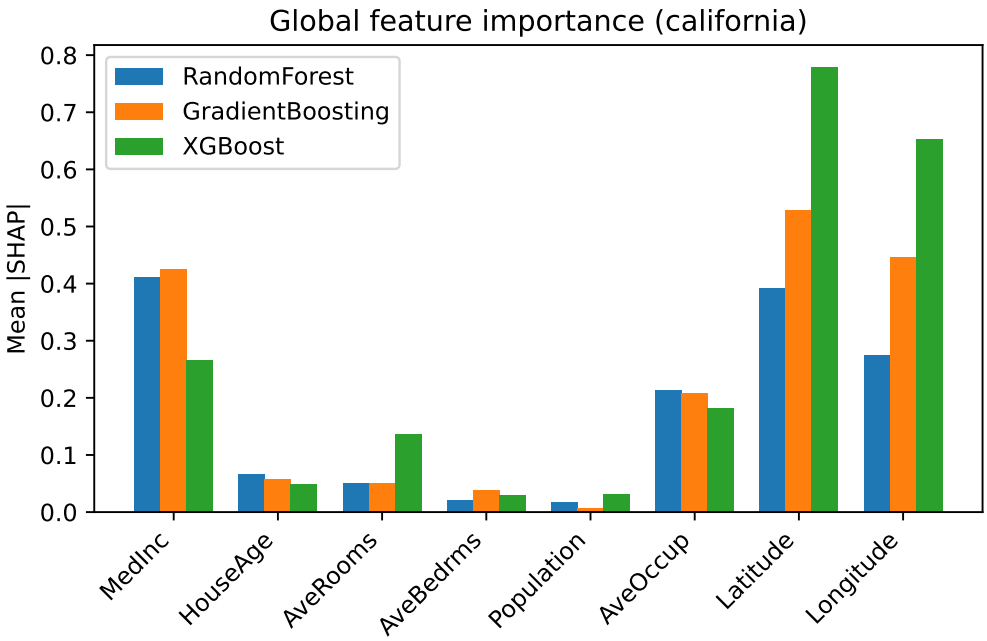

Figure 1: Mean |SHAP| per feature for California Housing (RF, GBM, XGB).

method. Reporting which model and which explainer were used, or checking agreement across models and across methods (SHAP vs. LIME), can make conclusions more reliable.

6 Conclusions

We quantified SHAP-based and LIME-based global feature importance stability across Random Forest, Gradient Boosting, and XGBoost on three tabular datasets (California Housing, Adult, Titanic), plus *instance-level* (local) stability and SHAP–LIME agreement.

Global vs. local SHAP. Pairwise Spearman correlation of global (mean |SHAP|) rankings was high (e.g., 0.88–0.98), but per-instance agreement was much weaker: only a minority of instances had mean pairwise $\rho \geq 0.9$ or identical top-k sets across all model pairs. So global stability can mask substantial disagreement at the instance level; conclusions drawn from global SHAP alone do not guarantee stable local explanations.

LIME agrees with itself; SHAP vs. LIME does not. LIME global rankings agreed perfectly across the three models (pairwise Spearman and top-k Jaccard both 1.0). That likely reflects LIME's design: it fits a *model-agnostic* local linear surrogate, so the same kind of linear summary can yield the same feature ordering across different underlying models. It does *not* mean the three models use the same features internally—SHAP, which is model-specific (tree structure), showed weaker agreement across models and often disagreed with LIME (SHAP–LIME rank correlation per model low or negative, e.g., -0.07 to -0.43). So perfect LIME-vs-LIME agreement is a property of the explainer, not evidence that models are equivalent; and the same model can be described as "these features matter" by SHAP and "those features matter" by LIME. Method choice drives the story as much as the model does.

Implications. A single model's explanation (SHAP or LIME) should not be treated as the only valid answer. Practitioners should: (i) report which model and which explainer were used; (ii) check agreement across models (e.g., global and, where relevant, local stability); (iii) check agreement across methods (e.g., SHAP vs. LIME) before drawing strong conclusions from feature attributions.

Future work could include multiple random seeds for confidence intervals and extension to

non-tree models.

Data Availability

Code and instructions to reproduce results are available in the project repository. Datasets are public (California Housing, OpenML adult and titanic).

References

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